Performance Analysis of a Hybrid ECC-LCG Lightweight Authentication Protocol for MQTT-Based IoT Networks

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1 Introduction: Motivation and Description of Problem

With the rapid growth in the number of devices connected through the Internet of Things (IoT), billions of resource-constrained devices have penetrated critical infrastructures like healthcare, smart cities, and industrial automation. According to [5], these devices are often deployed in unattended environments and hence are highly susceptible to security attacks like Man-in-the-Middle (MITM), replay, and impersonation.

The conventional cryptographic protocols (for example, RSA or AES-256) are computationally intensive, memory intensive, and energy intensive, which is beyond the capacity of low-power IoT sensor devices. Using these intensive protocols causes high latency and drains the batteries rapidly.

To solve this, lightweight cryptography (LWC) was proposed. [2] synthesizes several dozen papers to demonstrate that schemes are generally assessed in terms of their entropy, performance, energy, and throughput, with ECC generally outperforming RSA for comparable security levels. Survey- and cryptanalysis-style work, such as [5], discusses authenticated key exchange for IoT, specific protocol proposals, and common issues with schemes, such as a lack of forward or backward secrecy, offline guessing, and synchronization, which are all driving factors for clean, minimal round-trip authentication schemes. For smart-city IoT, for instance, LAID [4] presents a framework for lightweight, multi-phase authentication with ECC. For IoT with an MQTT focus, DLKS-MQTT [3] emphasizes the importance of dynamic, lightweight session material with LCGs and lightweight hashing. For industrial IoT, ECC-based authentication with formal analysis is similarly promoted [1]. These various research efforts demonstrate that there is still room for a unified template for IoT authentication, which addresses MQTT session establishment in particular, with DLKS and LAID aligned for their emphasis on ephemeral freshness, on the one hand, and structured ECC authentication, on the other.

Can a hybrid protocol, which implements both phased ECC mutual authentication using the LAID scheme and DLKS with LCG for seeding, reduce the computation, communication, and energy requirements of the handshake process compared to (i) the standard MQTT/TLS scheme and (ii) the standalone LAID scheme, in IoT environments with limited resources, without compromising on the resilience to replay, MITM, and desynchronization attacks? To answer this, a detailed sketch of the proposed protocol (Figure 1) needs to be provided, as well as an implementation or simulation of the proposed scheme to compare with the relevant metrics outlined in Section 2.2.

2 Possible Approaches

This project aims to explore a hybrid approach that is lightweight by using the concept of Dynamic Lightweight Key Sharing (DLKS) in conjunction with the four-phase LAID-style authentication model, with modifications for keys used in MQTT control and application data.

2.1 Algorithm Description

The proposed approach utilizes a Linear Congruential Generator (LCG) for fast ephemeral seeding and nonces to guarantee the freshness of the session, in the spirit of [3]. The seed is used for ECC-based scalar multiplication ($Q = s \cdot P$) and short message authentication tags in a certificate-free handshake, inspired by [4]. This is to minimize round trips with respect to PKI/TLS, but still maintain mutual authentication before MQTT CONNECT/PUBLISH.

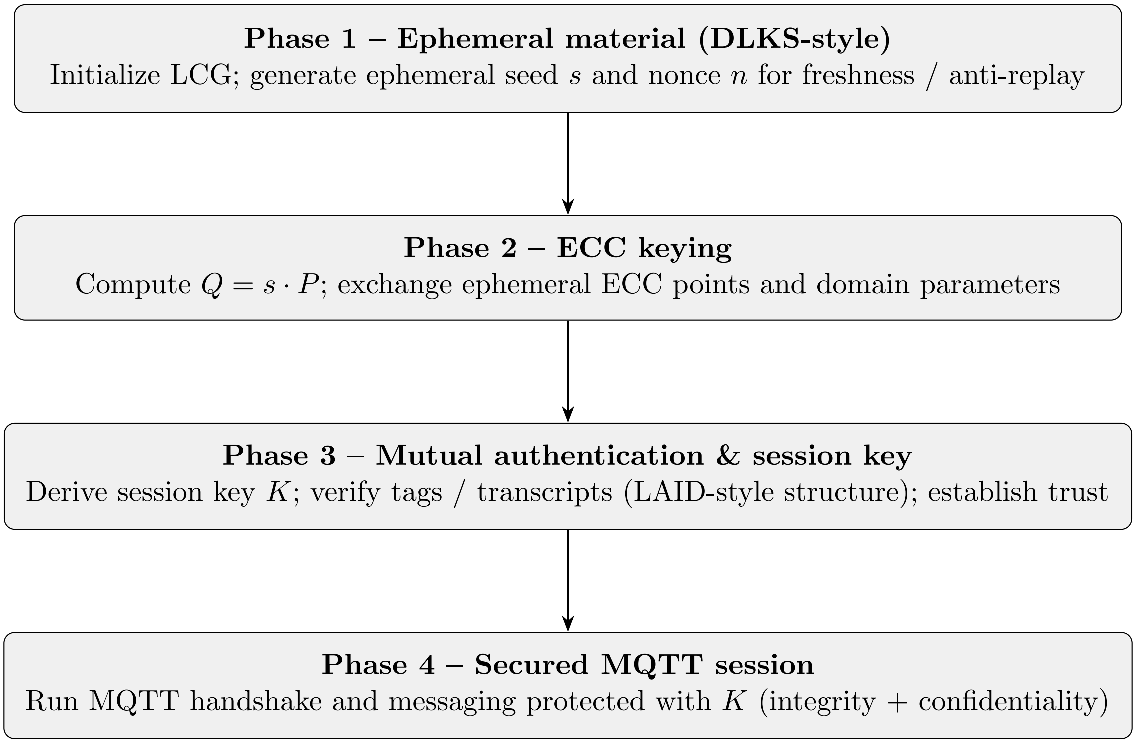

Figure 1: End-to-end pipeline at the level of this proposal (message fields and splits to be determined in Week 1).

2.2 Experiments to Conduct

I propose to perform a quantitative comparative analysis with two alternative evaluation paths to ensure that the results are compatible with both “constrained node” and “MQTT fidelity” requirements:

- *Network/device simulation:* Simulate networks/devices with Cooja (Contiki-NG) or NS-3, which can be used to evaluate multi-hop or dense IoT radio networks. In this case, we focus on CPU time per handshake, energy in mJ (from radio and MCU models), and bytes on the air.
- *Application benchmark (MQTT):* Perform an MQTT benchmark with a Mosquitto broker and Python or C clients to evaluate end-to-end handshake and first-publish latency for MQTT+TLS and the proposed protocol with the added layer above the socket, with similar metrics such as time, bytes, and optionally power proxy via timing of an IoT-class single-board device, if possible.

Baselines: MQTT-TLS (e.g., TLS 1.2/1.3 as provided by the stack), and standalone LAID-style ECC-based authentication [4] without the LCG/DLKS ephemeral layer. Metrics:

- Computation time (ms): Latency from request to completion of authenticated session (and first application message, if measurable).
- Energy consumption (mJ): Estimated per-handshake value from simulator or device energy models, if available.
- Communication overhead (bytes): Total handshake octets, including certificates or ECC points as necessary.

3 Research Plan

- Week 1 (April 7): Complete the structured notes on [2,5], including the exact message flow descriptions for Figure 1, and lock the parameters of the curve, the LCG implementation, and the threat model (in line with AVISPA/ProVerif-based claims).
- Week 2 (April 14): Stand up Cooja/NS-3 scenarios and Mosquitto/baseline TLS clients; implement reproducible measurements of the handshake.
- Week 3 (April 21): Implement the hybrid ECC-LCG layer; perform stress tests (various node densities/publish rates); gather raw logs.
- Week 4 (April 28): Evaluate the results (tables, graphs); compare with baselines; write the full technical report.
- Final presentation (May): Present the slides; discuss the limitations (e.g., verification scope, simulated vs. real-world deployment).

References

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[4] Aqeel Khalique, Farheen Siddiqui, Mohd Abdul Ahad, and Imran Hussain. Lightweight authentication for iot devices (laid) in sustainable smart cities. *Scientific Reports*, 15(1):25410, 2025.

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